

Lab 5

NO late submission. Grade = 0 after the deadline.

Due: 13:50h 29/10/2025

Submission:

- **Lab#_YOURNAME.ipynb**
- Must include all data file (**.csv**,...) and **.ipynb** in your submission
- **.csv** and **.ipynb** must be in the **same** folder (pd.read_csv('data.csv'))
- No use of LOCAL libraries, NO local directory for libraries

The `pivot_table()` function in pandas is a powerful tool for summarizing and aggregating data within a DataFrame. It allows restructuring data by specifying **index** (rows), **columns**, and **values** to be aggregated.

pandas.**pivot_table**(data, values=None, index=None, columns=None, aggfunc='mean', fill_value=None, margins=False, dropna=True, margins_name='All', observed=<no_default>, sort=True)

Use **'mtcars.csv'**

MUST USE `pandas.pivot_table(df,...)`

1. Group by **cyl**, then apply aggregate functions (**mean**) on **'mpg'**
2. Group by **cyl**, then apply aggregate functions (**mean**) on **['mpg', 'hp']**
3. Group by **[cyl, am]**, then apply aggregate functions (**mean**) on **['mpg', 'hp']**
4. Group by **[cyl]**, then apply aggregate functions (**mean, std**) on **['mpg', 'hp']**
5. Group by **[cyl, am]**, then apply aggregate functions (**mean, std**) on **['mpg', 'hp']**
6. Group by **[cyl, am]**, then apply aggregate functions (**mean, median, std**) on **'mpg'** and then apply aggregate functions (**min, max**) on **'hp'**. Also, use **fill_value = 0, margins = True, margins_name = "Total"**.
7. Group by **[cyl]**, set pivot table **column** as **[gear]**, then apply aggregate functions (**mean, std**) on **['mpg', 'hp']**. Also, use **fill_value = 0, margins = True, margins_name = "Total"**.
8. Group by **[cyl, am]**, set pivot table **columns** as **[gear, carb]**, then apply aggregate functions (**mean, std**) on **['mpg', 'hp']**. Also, use **fill_value = 0, margins = True, margins_name = "Total"**.

9. Group by [cyl, am], set pivot table **columns** as [gear, carb], then apply aggregate functions (lambda function: $\text{sqrt}(\text{sum}(x*x))$) on ['mpg', 'hp']. Also, use fill_value = 0, margins = True, margins_name = "Total".
10. Use df.apply() and a lambda function, insert a new column called "Code" that uses the FIRST 3 letters of the "model" and 2 letters of the code of the COUNTRY: {code: countrycode}. all LOWERCASE.

Model	Country Code
Toyota, Honda, Datsun, Mazda	jp
Merc, Ferrari, Volvo, Porsche, Maserati, Lotus	eu
Others	us

For example:

Mazda RX4 → {'maz': 'jp'}

Porsche 914-2 → {'por': 'eu'}

Dodge Challenger → {'dod': 'us'}